Date: Roll No.

III SEMESTER B.Tech (ECE, EIOT)

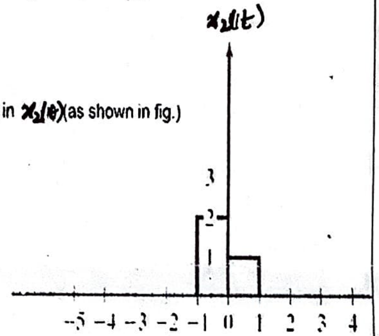
MID-SEMESTER EXAMINATION 2021

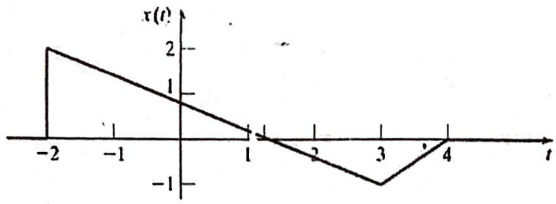
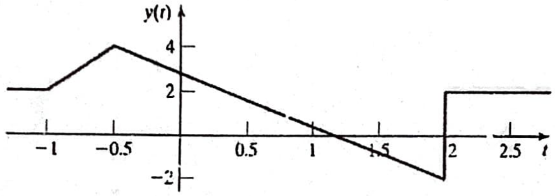
Course Code- ECECC05 / EIECC05

Course Title- Signals & Systems

Time: 1:30 Hrs Max Marks: 25

Attempt all questions. Missing data / information (If any) may be suitably assumed & mentioned in the answer.

Q. No.		Questions	Marks	CO
Q 1	a	You are given an LTI system. The response of the system to an input $x_1(t) = u(t) - u(t-1)$ is a function $y_1(t)$ What is the response of the system to the input in $x_2(t)$ (as shown in fig.) in terms of $y_1(t)$ 	03	CO2
	b	Suppose that $x(t) = 1 \quad 0 \leq t \leq 1$ $= 0$ elsewhere and impulse response of the system, $h(t) = x\left(\frac{t}{\alpha}\right) \quad 0 \leq \alpha \leq 1$ Determine and sketch $y(t) = x(t) * h(t)$.	02	CO2
Q 2	a	A discrete time periodic signal with time period N is applied as input to an LTI system with impulse response $h[n] = exp(-\lvert n \rvert)$. Is the output signal periodic? If yes what is the time period.	03	CO1 CO2
	b	Let h[n] be the impluse response of a causal and stable discrete time LTI system. Comment on the stability and causality of the cascade of 2 LTI sytems with impulse responses h[n] and h[-n].	02	CO2

Q 3	a	Express y(t) as a function of x(t).).  (a)  (b)	03	CO1
	b	The Energy of x(t) above is Ex, what is the energy of 2 y(t-2).	02	CO1
Q 4	a	Express $x[n] = [u[-n+3] - u[n]].[\delta[n+1] + \delta[n+4]]$ in terms of shifted impulse functions.	03	CO1
	b	Consider a discrete-time system whose input x[n] and out put are related as $y[n] = \sum_{k=-\infty}^{\infty} 2^{k-n} x[k+1]$. Is the system LTI?	02	CO1, CO2
Q 5	a	Let $x(t)$ be a periodic signal whose Fourier series coefficients are $X_n = 2, n = 0$ $= j\left(\frac{1}{2}\right)^{\|n\|}$, otherwise Find x(t).	03	CO3
	b	What are the fourier series coefficients of $\frac{dx(t)}{dt}$.	02	CO3

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